

CivicLens

Visual Fault-to-Civic-Action Platform

Google Cloud Rapid Agent Hackathon • June 2026 • Team Brief

"Turn every resident's phone into a civic accountability tool — from photo to filed complaint to social pressure in under 60 seconds, with zero manual effort."

What We Are Building

Most civic reporting apps stop at detection. CivicLens closes the loop — from visual intake to institutional action — automatically. A resident uploads a photo or video of a fault (pothole, garbage, broken infrastructure). A multi-agent pipeline powered by Gemini 3 VLM then classifies, contextualizes, routes, drafts, sends, and tracks — all without the user needing to figure out who to call, what to write, or how to follow up.

Hackathon Track	Open Ended + MongoDB + Elastic (multi-partner)
Core Technology	Gemini 3 VLM, Google Cloud Agent Builder, MCP servers
Target Users	Residents (reporters) + Municipal officers (dashboard)
Deadline	June 12, 2026
Key Differentiator	First platform to close the full loop: image → action → accountability

Multi-Agent Architecture

Each agent has a single clear job. Together they form a pipeline that takes a photo as input and produces a sent complaint, a social post, and a tracked case as output.

Agent	Name	Responsibility
Agent 1	Vision Agent	VLM analyzes image/video. Classifies fault type (road damage, garbage, infrastructure), estimates severity, generates structured fault description.
Agent 2	Context Agent	Fetches user name, address, phone. Pulls GPS coordinates, timestamp, street address via reverse geocoding. Packages full situational context.

Agent 3	Authority Lookup Agent	Based on fault type + location, identifies the specific responsible department (not just a generic office). Finds email, phone, working hours, distance, travel time.
Agent 4	Communications Agent	Drafts formal complaint email with all context pre-filled. Generates social media post with location, date, severity, photo caption. User just taps send.
Agent 5	Tracking Agent	Assigns case ID, logs to MongoDB. Monitors resolution. If unresolved after X days: auto sends follow-up email and re-generates social post.

End-to-End User Flow

1. Resident opens app, uploads photo or video of fault
 - VLM classifies: fault type, severity level, visual description
 - GPS, user profile, and timestamp automatically fetched
2. System identifies the right authority
 - Specific department matched (Roads / Sanitation / Utilities)
 - Contact email, working hours, distance shown in-app
 - Google Maps deep link generated for navigation
3. One-tap action suite
 - Complaint email pre-drafted — user reviews and taps Send
 - Social post auto-generated with location, date, photo, caption
 - Case ID issued, report logged
4. Accountability loop
 - Status tracked against case ID
 - Auto follow-up if unresolved after threshold days
 - Escalation documented and re-posted on social if needed

Advanced Features — What Makes This Win

Issue Deduplication If multiple residents report the same fault, reports are bundled. The complaint reads “14 residents affected” — that number changes how authorities respond.	Severity-Based Routing Fault type determines the specific department, not a generic inbox. Pothole → Roads. Garbage → Sanitation. Broken light → Utilities.
Auto Follow-Up Loop Unresolved after X days triggers automatic follow-up email and re-generated social post. Closes the accountability loop.	Authorities Dashboard Second interface for municipal officers: live heatmap of reports, prioritized by severity and volume. Makes the platform a B2G tool, not just B2C.
Offline-First Queue	Historical Pattern Detection

Poor connectivity is worst exactly where infrastructure problems are worst. Queue reports offline, auto-submit on reconnect.	MongoDB query checks if same location was reported before and unresolved. Agent escalates automatically with prior history attached.
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MCP Partner Stack

MongoDB MCP	Incident logging, deduplication, historical pattern queries, case tracking
Elastic MCP	Search across historical reports, fault clustering, geographic analytics
Gemini 3	VLM fault analysis, classification, severity estimation, text generation
Google Cloud Agent Builder	Multi-agent orchestration, deployment, managed pipeline
Google Maps API	Reverse geocoding, authority distance, navigation deep links

3-Minute Demo Arc

The demo must show: user command → agent plans → agent acts. No chatbots. Every step produces a real output.

<i>Step 1 — Upload photo of a pothole. Agent classifies: Road Damage, Severity High. GPS pin shown on map.</i>
<i>Step 2 — MongoDB query: Same location reported 2 months ago, unresolved. Agent auto-escalates. This is the surprise moment.</i>
<i>Step 3 — Email drafted and sent to Roads Dept. Social post generated. Case ID issued. All in under 60 seconds.</i>
<i>Step 4 — Switch to Authorities Dashboard. Live heatmap. 12 open cases ranked by severity. This is the scalability visual.</i>

Why This Wins

- Not a chatbot — a multi-agent system that takes real actions
- Solves a problem every judge has personally felt or seen
- Demo is self-contained, visually dramatic, and failure-proof
- Two interfaces (resident + authority) tell a complete scalability story
- Multi-partner MCP integration is natural and deep, not forced
- Deployable end-to-end in 10 days — no dependency on mocked data